

Koide Amplitude $d/c = \sqrt{2}$ from T^4/Z_3 Geometry

Derivation from the Z_3 Equal-Distribution Condition and the Galois Coupling Matrix
(Doc. 348)

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Status Markers

- [K]** *Confirmed* — derived from ξ , numerically verified.
- [B]** *Proven* — algebraically demonstrated.
- [S]** *Speculative* — open.
- [Q]** *Source* — external primary source, verified here.

Abstract

The Koide condition $Q = 2/3$ is equivalent to the amplitude condition $d/c = \sqrt{2}$ in the Z_3 circulant mass operator (Docs. A110, 352). This document derives $d/c = \sqrt{2}$ from the T^4/Z_3 geometry without using Koide or the lepton masses as input.

The key is the trace bilinear form of $GF(27)^*$ (Doc. 348 Theorem C): the coupling matrix between the neutrino orbits (f_1) and the lepton orbits (f_3) has normalised entries $|M_{ij}^{(13)}| = 1/\sqrt{2}$ at all non-zero positions. This is a consequence of the Z_3 equal-distribution: each element couples to exactly two of three partners, and the normalisation of the coupling strength forces the weight $1/\sqrt{2}$ per connection.

The same condition appears in the mass operator as $d/c = \sqrt{2}$: the circulant parameter d is the coupling weight between the isotropic part c and the Z_3 interaction term, and the equal-distribution forces $d/c = \sqrt{2}$.

Main findings: (1) $d/c = \sqrt{2}$ from Z_3 equal-distribution **[B]**. (2) Connection to the $GF(27)^*$ coupling matrix $|M^{(13)}|_{\text{norm}} = 1/\sqrt{2}$ **[B]**. (3) PDG satisfies $d/c = \sqrt{2}$ to $0.10 \cdot \xi$ **[K]**. (4) Bare FFGFT residual $+17 \cdot \xi$ explained by ε_i (101 %) **[K]**. (5) Open R113-bridge (origin of falling ε_i structure) **[S]**.

1 Starting Point

The Koide Amplitude Condition

The Z_3 circulant mass operator (Doc. A110) writes the square roots of the lepton masses as:

$$\sqrt{m_k} = c + d \cdot \cos\left(\theta + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2. \quad (1)$$

From $\sum_k \sqrt{m_k} = 3c$ and $\sum_k m_k = 3c^2 + \frac{3}{2}d^2$ it follows **[B]**:

$$Q \equiv \frac{\sum m_k}{(\sum \sqrt{m_k})^2} = \frac{1}{3} + \frac{d^2}{6c^2}. \quad (2)$$

The Koide condition $Q = 2/3$ is equivalent to **[B]**:

$$\boxed{d = c\sqrt{2}}. \quad (3)$$

This condition does not contain θ — Q depends exclusively on d/c (Doc. 352 orthogonality theorem **[B]**). The phase condition $\theta = 2/9$ is anchored in the geometry **[K]**; the amplitude condition $d/c = \sqrt{2}$ was previously open **[S]**.

Known: $1/\sqrt{2}$ in the Galois Coupling Matrix

Doc. 348 Theorem C derives from the trace bilinear form $\text{Tr}_{\text{GF}(27)/\text{GF}(3)}(g^{k+j})$ the coupling matrix between neutrino orbits (f_1) and lepton orbits (f_3) **[B]**:

$$M^{(13)} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad |M_{ij}^{(13)}|_{\text{norm}} = \frac{1}{\sqrt{2}} \quad \text{at all non-zero entries.} \quad (4)$$

The weight $1/\sqrt{2}$ follows from the $\text{GF}(27)^*$ orbit structure. The connection to the Koide amplitude condition is the subject of this document.

2 Derivation of $d/c = \sqrt{2}$

Step 1: The Z_3 Equal-Distribution Condition

The Z_3 symmetry of the torus T^4/Z_3 arranges the three lepton generations as equivalent elements of an orbit. A coupling between two Z_3 orbits (e.g. neutrinos f_1 and charged leptons f_3) has the property, via the trace bilinear form, that each element couples to exactly two of three elements of the other orbit (the coupling matrix (4) has exactly one zero entry per row).

The ℓ^2 normalisation of the coupling strength per row gives:

$$\sum_{j: M_{ij} \neq 0} |M_{ij}|^2 = 1 \implies 2 \cdot |M_{ij}|^2 = 1 \implies |M_{ij}| = \frac{1}{\sqrt{2}}. \quad (5)$$

This is the weight $1/\sqrt{2}$ from (4) **[B]**.

Step 2: Translation to the Circulant Mass Operator

In the circulant mass operator (1), c is the isotropic part (mean of $\sqrt{m_k}$) and d is the anisotropic part (amplitude of the Z_3 modulation).

Parameter identification:

- The isotropic part c corresponds to the self-coupling of an orbit — it sets the mass scale.
- The anisotropic part d corresponds to the coupling between orbits — it describes the interaction.

The weight of this coupling is fixed by the equal-distribution. The coupling matrix (4) normalised to the isotropic scale: every non-zero coupling has weight $1/\sqrt{2}$ [B].

Therefore the ratio of the anisotropic to the isotropic part is:

$$\frac{d}{c} = \frac{1}{|M_{ij}|_{\text{norm}}^{-1}} = \frac{1}{1/\sqrt{2}} = \sqrt{2}. \quad (6)$$

This is $d/c = \sqrt{2}$ [B].

Step 3: Geometric Reading

The three lepton generations form a Z_3 orbit; the three neutrino generations form a second. The trace bilinear form of GF(27) describes how these orbits couple to each other. The result is structurally forced:

1. The Z_3 symmetry allows each element exactly 0, 1 or 2 non-zero couplings to the other orbit.
2. The trace bilinear form selects exactly 2 (Doc. 348 Theorem C [B]).
3. The ℓ^2 normalisation forces the weight $1/\sqrt{2}$ per coupling.
4. In the mass operator this weight appears as $d/c = \sqrt{2}$.

The condition $d/c = \sqrt{2}$ is therefore not a fit to the measured value, but a consequence of the Z_3 equal-distribution in the T^4/Z_3 geometry [B].

3 Numerical Verification

Source	d/c	$d/c - \sqrt{2}$	Units ξ	Status
PDG [Q]	1.41420	-1.0×10^{-5}	$-0.098 \cdot \xi$	[K]
FFGFT bare	1.41649	$+2.28 \times 10^{-3}$	$+17.1 \cdot \xi$	[K]
FFGFT + ε_i	1.41422	$+8 \times 10^{-6}$	$+0.06 \cdot \xi$	[K]
Target (geometry)	$\sqrt{2} = 1.41421$	0	0	[B]

PDG satisfies the condition to $0.10 \cdot \xi$. The bare FFGFT residual of $+17 \cdot \xi$ is fully explained by the generation-dependent corrections ε_i (Doc. 352 §9, 101 %) [K]. PDG masses: $m_e = 0.51100$ MeV, $m_\mu = 105.658$ MeV, $m_\tau = 1776.86$ MeV [Q] (PDG 2024).

4 Status Overview

	Statement	Status
1	$Q = 2/3 \iff d/c = \sqrt{2}$ algebraically from Z_3 circulant	[B]
2	Q does not contain θ ; d/c and θ are orthogonal parameters	[B]
3	$d/c = \sqrt{2}$ from Z_3 equal-distribution of coupling to 2 of 3	[B]
4	Connection to $\text{GF}(27)^*$ coupling matrix $ M^{(13)} _{\text{norm}} = 1/\sqrt{2}$	[B]
5	PDG satisfies $d/c = \sqrt{2}$ to $0.10 \cdot \xi$	[K]
6	Bare FFGFT residual $+17 \cdot \xi$ explained by ε_i (101 %)	[K]
7	Dynamical derivation of the falling ε_i structure	[S]

Statement 3 closes the open bridge from Doc. 352 and register Doc. 190. Status: raised from **[S]** to **[B]**.

5 Note: Derivation Chain for Precise Lepton Masses

The lepton ladder is not in the chain for the precise mass derivation.

The lepton ladder ($m_i = r_i \cdot \xi^{p_i} \cdot \nu_i$, Doc. A100) yields masses at the percent level (1 %). It is a coarse framework, but not a necessary link in the precise derivation. The **[S]** entry in Doc. 352 §7 ("Koide is not contained in the FFGFT lepton ladder") concerns the ladder alone — it is correct and remains **[S]**. It is, however, *not* in the chain for the precise masses.

The chain for the precise lepton masses runs exclusively through the Z_3 circulant:

1. $\theta = 2/9$ from the Z_3 circulant geometry **[K]** (Docs. A110, 291)
2. $d/c = \sqrt{2}$ from the Z_3 equal-distribution condition **[B]** (this doc.)
3. Reference point m_μ/m_e **[Q]** — one declared empirical anchor (Doc. A110, P42)
4. $m_\tau = 1776.9690 \pm 0.0001$ MeV **[K]** (Docs. A110, A210)

This chain contains *no* **[S]** entry. Doc. A110 has no open points in the circulant derivation. All **[S]** entries in Doc. 352 concern residual analyses (ε_i structure, R113-bridge) or the ladder–Koide relation — none of these is in the derivation chain for the precise masses.

Conclusion: The derivation chain for the precise lepton masses is fully closed with the completion of $d/c = \sqrt{2}$ **[B]** (this document). No **[S]** blocks it.

6 Verification Script

- Numerical verification of the Z_3 normalisation condition, coupling matrix entries (4), d/c calculation from PDG and bare FFGFT masses, ε_i correction: `python/Dok353_Skripte/pruef_353_koide_amplitude.py`

7 Open Points

- Why do ε_i exhibit a falling generation-dependent structure ($\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$)? Attribution to QED extraction and SI projection is **[S]** (R113-bridge, Doc. 352 §10).
- Full derivation of CKM/PMNS mixing angles from the Galois bilinear form (Doc. 348 Theorem D) **[S]**.

- Are c and d individually fixed by the geometry, or only their ratio? **[S]**

8 Use of AI Tools

This document was produced with the assistance of AI-based tools. The questions, postulates and all content decisions originate with the author; the tools formulate, compute and generate verification scripts. Responsibility for content and errors lies entirely with the author. The detailed wording is in A010.